SMART SKINCARE

Find products that truly match your skin.

Foundation of Every Routine

Remove makeup, oil, and impurities without stripping essential moisture.

Hydrating Lip Care Essentials

Lightweight tints, luminous glosses, and hydrating balms with skincare benefits.

Discover New Brand Launch

Explore innovative formulas powered by trending ingredients and research.

DH, Emily, Jungmoon, Wonjoon

Background

Goal

Build an interactive skincare recommender that suggests products based on a user's skin type and ingredients

Planned Outputs

- Web app for user input
- Recommendation engine

Use Case

Personalized Product Recommendation

- User selects skin type
- System recommends matching products
- Filters: category, brand, price, exclusions
- Product pages show ingredients, benefits, irritants.

Data Sources

Cosmetic Ingredients

- ~1,500 skincare products
- Includes detailed ingredient lists
- Organized by product categories
- Labeled by skin type (dry, oily, combination, etc.)

Skincare Product Reviews

- 2,000+ skincare products
- Includes user reviews and ratings
- Provides real-world feedback on product performance

Data Manipulation

Ingredient Standardization

- Parsed raw ingredient lists from product datasets
- Standardized ingredient names using Paula's Choice
- Resolved ingredient synonyms with PubChem and INCI Decoder

Ingredient Knowledge Mapping

- Linked ingredients to different skin types
- Based on Paula's ingredient benefits and functions
- Expanded with curated ingredient mappings

How the Model Scores Products

Input

- User profile
- Product ingredients
- Ingredient knowledge base
- Product ratings and reviews

User Profile

- Dry, Oily, Sensitive, Pigmentation, & Wrinkles
- Each factor is scored from 0 to 1

Scoring Process

- Add points for ingredients that help the user’s skin needs
- Subtract points for ingredients that may be harmful or irritating

Recommendation System

Penalties

- Lower score if the Paula's Choice rating is poor
- Extra penalty for high-risk ingredients
- AHA/BHA for sensitive or very dry skin
- L-ascorbic acid for sensitive skin

Final Adjustment

- Uses reviews to make results more balanced and reliable

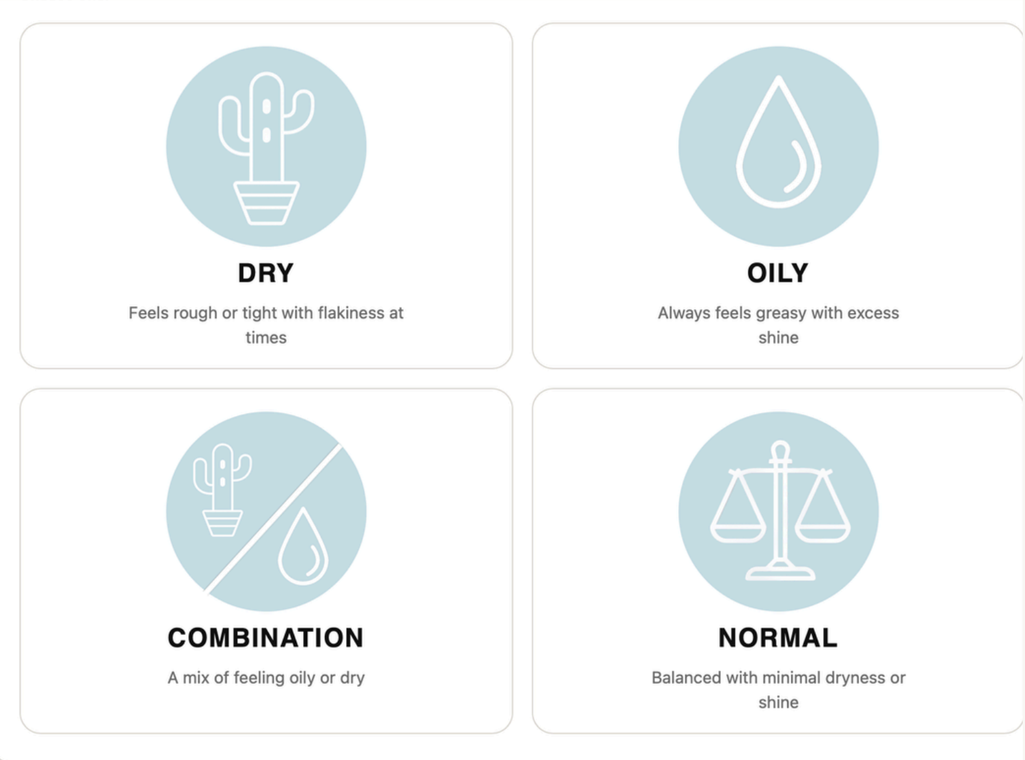
Final Score = Ingredient Score − Penalties + Ratings + Reviews

- Normalized to a 0–100 scale
- Higher score = better match for the user's skin needs

$$\text{norm}_i = 0 + (s_i - s_{\min}) \times \frac{100 - 0}{s_{\max} - s_{\min}}$$

Design

1. **User takes the skin test**
   - Skin type + concerns
2. **System saves results**
   - Stored in user profile
3. **System pulls products**
   - From product database
4. **System scores products**
   - Based on ingredients + user profile
5. **UI shows recommendations**
   - Name, brand, ingredient notes, score



Demo

Home

SMART SKINCARE

Find products that truly match your skin.

The skincare market is growing rapidly, with an overwhelming number of products available. While more choices can be helpful, they often make it difficult for consumers to know which products are right for their skin. The best way to choose skincare is by understanding product ingredients and how they match your skin type, but this can be challenging for most people.

Our app simplifies this process. After a short skin type survey, we analyze skincare product ingredients and recommend up to 30 products that best match the user's skin type. Users can then easily explore and purchase them through platforms like Sephora, TikTok Shop, or Amazon.

Take the Skin Test

Featured Collections

Lesson Learned

- Git workflow (branching, PR, code review) improved collaboration
- Clear documentation (README) helped project usability
- Virtual environments and pylint improved code quality

Future Work

- Expand automated testing
- Improve project packaging and structure
- Add contribution guidelines and issue templates

Thank you!

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